GLENCORE

Data Standard
General Configuration

12

Document properties

Date, Version	4/05/2022 v1.0
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Comment/s	Published

Contents

1. Introduction....4
1.1 Master Data Standards - Purpose....4
1.2 General Configuration Standard – Purpose....4
1.3 Using this document....4
1.4 Governance....5
2. Introduction to General Configuration Requirements....6
2.1 General....6
2.2 Master Data Object Naming....6
2.3 Short Text Naming....6
2.4 Common Requirements Across All Defined Data Objects....7
3. Master Data Objects Items....8
3.1 Planner Group....8
3.2 Revisions....8
3.2.1 Revision – Weekly....8
3.2.2 Revision – Shutdown....8
3.2.3 Revision – Work Order Grouping....8
3.3 Document Numbering....9
3.3.1 Equipment Document Numbering Standard....9
3.3.2 Work Order Numbering Standard....9
3.3.3 Task List Numbering Standard....10
4. Short Text Items....11
4.1 Functional Location Short Text Description....11
4.2 Equipment Short Text Description....11
4.3 Task List Header Short Text Description....11
4.4 Task List Operation Short text Description....12
4.5 Maintenance Item Short Text Description....12
4.6 Maintenance Plan Short Text Description....13
4.7 Measuring Point Short Text Description....13
4.8 Construction Type Short Text Description....14
4.9 PM Assembly Short Text Description....14
4.10 Work Centre Short Text Description....15
4.10.1 Work Centre (002 – Person)....15
4.10.2 Work Centre (001 – Machine)....15
4.11 BOM Item Text 1 and 2....16
Appendices....17
Appendix 1 - Restricted Characters....17

Appendix 2 - Frequency codes 17
Appendix 3 – Discipline Codes 18
Appendix 4 - Activity Code Abbreviations 18
5. STAKEHOLDER SIGNOFF 20

1. Introduction

1.1 Master Data Standards - Purpose

The Data Standard provides clear and prescriptive definitions and rules that support a consistent and common definition of data entities across all business systems.

It describes:

- What rules / guidelines are applied to determine how many of the relevant data entity will be created.
- What rules will be applied to determine the codes used to create the relevant data entity (e.g. Plant code or Material code);
- What rules will be applied to determine the name / short description of the relevant data entity (e.g. Plant name or Vendor name);
- Rules for the use of views / sub-entities within the Data Entity;
- Identification of the key Attributes on the data entity whose usage needs to be standardised;
- When multiple values are available for the nominated 'key' attributes, the rules definition should list the criteria to use to determine the appropriate value; and
- If applicable, any rules / guidelines for the association of the relevant data entity with other data entities.

The Data Standard is **NOT** used to document:

- 'Technical' documentation on how the system has been configured and the design rationale for the configuration
- Identifying the operational need on when these Standards should be applied, They are to provide the guidance on how the Standard is applied
- List of actual values for the data entity / attributes within the data entity except where it is required to explain the application of the data standard
- User training or procedural documentation for master data maintenance
- Data migration

1.2 General Configuration Standard – Purpose

The purpose of this Standard is to:

- Identify the minimum Master Data field requirements for SAP General Configuration at Glencore
- Ensure alignment with the Solution Building Block SBB_D90 Maintenance Master Data and Asset Management Framework Elements 5, 8 and 12
- Provide the guiding principles to manage a General Configuration Master Data
- Identify the primary audience for the application of this standard

1.3 Using this document

The Data Standard is targeted at the following audience:

- Master Data maintenance teams in order to understand how the object needs to be maintained
- End users supporting the maintenance function whereby appropriate knowledge is required to use the system effectively which include but are not limited to; Planners, Schedulers, Reliability Engineers etc.
- Business Intelligence team to assist with the design of reporting solutions
- Project teams whose scope impacts the data object to ensure the solution design is compliant with the approved data standard
- Transition to operations teams to assist in the development of user guides

1.4 Governance

Master Data Governance (MDG) is the governance tool used to create, maintain and retire Master Data objects in Plant Maintenance. The Asset Management group provides the Governance for all requirements described in this Standard as well as Approval of any change to related Master Data.

This Standard shall be reviewed as often as the business requires but at a minimum of every five years.

The Standard is subject to Glencore's Continuous Improvement (CI) processes based on Glencore's shared experiences and Lessons Learned.

Revisions, reviews, and updates to the Standard take place under the guidance and approval of Regional Asset Management and Subject Matter Expert (SME) leaders.

2. Introduction to General Configuration Requirements

2.1 General

The design for Master Data Objects is defined in the master data standards, which includes design intent along with creation, change & deletion requirements. They also include object field entries, mandatory and optional details, relevant tables and important interactions with other data objects.

The General Configuration standard has been developed to provide further guidance on the following:

- Master Data Object Naming
- Short Text Naming

2.2 Master Data Object Naming

The requirements for naming of master data object fields are not defined in the master data design (MDD) standards applicable to the object where developed. Naming is instead provided with reference to the Plant Maintenance (PM) Configuration Tables whereby it is either classified as a Global standard, or specific to the Region in which it is used, i.e. Australia, North America etc.

Naming for Global master data objects is contained within the following document:

- PM_Global_Master_Data_Configuration_-_Glencore_-_v*XX**_*date** (version & date fields continually updated)

Global master data object naming is used for the following fields as an example:

- Maintenance and Planning Plants
- Object Type
- Priorities
- User and System Status

Naming for Region specific master data objects is contained within the following:

- PM_*YYY**_Master_Data_Configuration_-_Glencore_-_v*XX**_*date** (version & date fields continually updated)

Whereby YYY* is defined as the Region being Africa (AFR), Australia (AUS), North America (NAM), South America (SAM).

Region specific master data object naming is used for the following fields as an example:

- Document Numbering
- FLOC (Functional Location)
- Planner Group
- Revisions
- Task List Groups
- Work Centres (Person and Machine)

This standard will provide information on master data objects which have not been defined in the suite of MDD's only. Naming from these documents will inform Short Text Naming.

2.3 Short Text Naming

The requirements for short text fields are not defined in the master data design standards applicable to the object. Short text fields within Master Data objects are not validated through table entries hence this configuration standard defines free text object short text fields with respect to:

- How it should be used
- Intent of the field
- The required format
- Any sequencing requirements
- Allowable abbreviations and
- Punctuation and special character requirements.

This data standard summarises and clarifies the creation of short text to provide direction and assist the business in complying with data standards for individual objects. This configuration standard should provide a consistent standard on short text descriptions across a number of master objects to ensure that SAP search capability can be fully utilised and information readily recognised by end users.

Short text requirements have been defined for the following objects:

- Functional Location
- Equipment
- Maintenance item
- Maintenance Plan
- Measuring point
- Task list header
- Task list operation
- Construction type and PM Assembly
- Work Centre
- BOM and Alternative Text

2.4 Common Requirements Across All Defined Data Objects

The following items should be applied to all data objects defined above:

- The short text should be used to describe the object or activity that is being defined
- Upper case shall be used at all times (Equipment is defined by Master Data Governance (MDG) Taxonomy derived from Equipment Classification and Characteristics table)
- Complete words or numbers should be used in preference to abbreviations, where space permits
- Illegal characters have been defined & shall not be used (Appendix 1)
- All abbreviations & words will be separated by a space
- There are no restrictions on long text use

3. Master Data Objects Items

3.1 Planner Group

A Planner Group is:

- Used to group work that is responsible to plan and/or execute by work
- Set at an Equipment or Task List to auto populate on a Work Order

The Planner Group field on Corrective Work Orders should be updated as required. Refer to Region specific configuration tables for Planner Group naming.

3.2 Revisions

At Glencore Copper there are 3 types of Revisions used:

- Weekly
- Shutdown
- Work Order Grouping

Revisions are created via an MDO request and will follow the following standards.

3.2.1 Revision – Weekly

A weekly revision defines the week of year and is configured Monday to Sunday in the format:

- Revision - WK-<Week of Year>-<Year>, i.e. WK-36-21
- Revision Description – <Week of Year>, <Year>. i.e Week 36, Year 2021
- Revision Start Date & Time - <Start of Week>, i.e. 06.09.2021 00:00:00
- Revision End Date & Time - <End of Week>, i.e. 12.09.2021 23:59:59

3.2.2 Revision – Shutdown

A Shutdown revision is used to group work for a specific plant shutdowns and will follow this format:

- Revision – <Location><*Department/Asset><Year><Sequential Number>, i.e. CSCV2101. Refer to Region specific configuration tables.
- Revision Description – <Location><*Department/Asset><Month Year>. i.e COPPER SMELTER CONVERTOR SHUT AUGUST 2021
- [+]Revision Start Date & Time - <Start of Shutdown>, i.e. 06.09.2021 00:00:00
- [+]Revision End Date & Time - <End of Shutdown>, i.e. 12.09.2021 23:59:59

Note:

1. *Select the primary driver of the shutdown as the Department
2. *Department Code can be created by Planner (i.e. not part of MDG)
3. [+]Not Mandatory

3.2.3 Revision – Work Order Grouping

A Work Order Grouping revision is used to group work for a specific plant and/or type e.g. Structual Work.

- Revision – <Location><Department/Asset><Reason><Sequential Number>, i.e. CSCVST01 See table below
- Revision Description – <Location><Department/Asset><Reason><Year>. i.e COPPER SMELTER CONVERTOR STRUCTUAL WORK 2021

- Revision Start Date & Time – Not Required for Grouping Revisions
- Revision End Date & Time - Not Required for Grouping Revisions

3.3 Document Numbering

A Documents Numbering system shall be adopted for the following documents:

- Equipment
- Work Orders
- Task Lists

In each instance, the numbering system shall articulate the document numbering to a maximum of 25 characters.

3.3.1 Equipment Document Numbering Standard

An equipment document numbering standard shall be used to identify equipment documents from other document types and enable searching of documents by site, document type and SAP object (Equipment or FLOC).

An Equipment document shall be created and linked to the SAP object to store historical information such as; Reports, Certificates, Specifications etc. Equipment documents related to a specific work order can be linked to both SAP objects equipment and work order.

This naming standard **does not** apply to any equipment documents which are held by the local site Technical Records department having their own numbering system, an example being engineering drawings. All Technical Records documents shall be linked to the SAP object.

Equipment (XX-XX-XXXXXXXXXXXXX-XXXX):

- Segment 1: Site Prefix (2 characters, refer to Region specific configuration tables)
- Segment 2: Document Type (3 characters, refer to Region specific configuration tables)
- Segment 3:
    - Tech ID for Single Equipment (up to 13 characters)
    - MLP plus Level 4 or 5 FLOC for Multiple Equipment (up to 10 characters, being MLP-XXXXXX)
- Segment 4: Sequential Number (3 characters)

Example:

- IS-CAL-ILZ055296-0001 (Single)
- IS-CAL-MLP-MILSAG-9999 (Multiple)

3.3.2 Work Order Numbering Standard

Work order document numbering standard shall be used to identify work order documents from other document types and enable searching of documents by site and work order.

Work order documents are created specifically to store an electronic copy of the completed work order with any hand written comments or completion sheets for historical purpose.

Work order documents shall be linked to the SAP object work order only and not to the equipment. Reports associated to a work order should be created as an equipment document and linked back to the work order Equipment.

Work Order (XX-XXXXXXX-XXXX):

- Segment 1: Site Prefix (2 characters)
- Segment 2: Work Order Number (nominally 7 characters)
- Segment 3: Sequential Number (4 characters)

Example:

- CS-10005623-0001

3.3.3 Task List Numbering Standard

Task list document numbering standard shall be used to identify task list documents from other document types and enable searching of documents by site and task list type.

Task list documents are used to link specific task list instructions to a task list header or operation/s.

A Task list document number does not contain the task list group, counter or operation and intead users a user defined sequential number of 9 characters

Task List (XX-XXX-X-XXXXXXXXX):

- Segment 1: Site Prefix (2 characters)
- Segment 2: Document Type (3 characters) being TSK
- Segment 3: Task List Type (1 character)
    - A – General Task List
    - E – Equipment Task List
- Segment 4: Sequential Number (9 characters)

Example:

- IS-TSK-A-000101000

4. Short Text Items

4.1 Functional Location Short Text Description

The short text of Functional Locations should describe the unique Functional Location such that there is no confusion when choosing a Functional Location. It shall describe the Functional Location in detail with a maximum of 40 characters.

Sequence in which information will appear:

- Meaningful Text e.g. TRUCKS - DUMP

Acceptable:

- TRUCKS - DUMP
- CONCENTRATOR PRIMARY CRUSHING
- VILLAGE ACCOMODATION ROOMS

Not Acceptable:

- Underground truck

4.2 Equipment Short Text Description

- Shall describe the Equipment in detail (up to 40 characters)
- In MDG the description is derived from the Tech ID field, Equipment classification description followed by mandatory Characteristic values

Sequence in which information will appear:

- Meaningful Text e.g. IMELD44,LOADER;8000 LB

Acceptable:

- IGFDB494,DISTRIBUTION BRD
- IME8922,PUMP
- IMEDTK13,TRUCK MINING;50 TON,760 HP

4.3 Task List Header Short Text Description

The Task List header description will be used to help identify both routine and non-routine tasks. This text should be used to identify the tasks and the type of object the tasks apply to. The assembly assigned to the task list header will be used to identify the relevant Maintainable unit or maintainable function.

- Task List description shall begin with the frequency and unit of measure of the package that has been assigned in the Task List operation where applicable. (Appendix 2). If the Task List is for an adhoc task, the frequency does not need to be applied.
- Discipline codes (Appendix 3) can be an optional addition to help distinguish between tasks
- Activity codes can be used to identify the type of the activity (See Appendix 4). If there is no frequency then the discipline code will be the first item in the task list description (The data standard has identified how task lists with no strategy can be identified)
- Any additional detail required will be within the long text
- General Task Lists must not reference specific master data objects or Tech IDs.
- Equipment or FLOC Task Lists may reference a specific equipment or FLOC
- Shall describe the task in detail (up to 40 characters)

Information build for Task List Description

Sequence in which information shall appear:

- Frequency Code e.g. 3M (3 monthly)
- Discipline Code e.g. MECH
- Activity Code e.g. INSP (Inspect)
- Object Identifier e.g. CRUSHER

Acceptable:

- 3M MECH INSP CRUSHER
- 2Y INST CALB CONVEYOR WEIGHTOMETER
- 500H LUBE SVCE MINING TRUCK
- MECH REP PUMP (Adhoc)

Unacceptable:

- 02M MECH INSP, CV001
- 10,000H Lube Haul Truck

4.4 Task List Operation Short text Description

Task list operations shall summarise the required activities that each work centre will perform. Long text within the operation shall be used if the activity requires further description. Details of the actual work activities shall be contained within work instruction documents that will be attached to the operation and will be printed with the work shop papers. Multiple operations can exist within the task list and the operation text is transferred to the work order operation text when a task list is used to populate a work order.

- Multiple operations shall not be used to detail the individual work activity steps or work procedures, any work procedures shall be contained within work instructions attached to the operation
- The first word shall be a verb describing the trades task to be completed and will not be abbreviated
- Long text shall not be used to define detailed work instructions. Detailed procedures are to be attached to the Operation via document link to a controlled document system repository.
- General task lists shall be used so the operation short text shall have no reference to specific master data objects or Tech ID's
- Shall describe the task list operation in detail (up to 40 characters)

Acceptable:

- INSPECT CRUSHER
- CHANGE OUT CYLINDER
- SERVICE HAUL TRUCK

Unacceptable:

- 02M MECH INSP, CV001
- 10,000H Lube Haul Truck

4.5 Maintenance Item Short Text Description

The maintenance item short text descriptions transfer to the maintenance order short text and should replicate the task list header description unless by exception. The maintenance item short text description shall:

- Commence with the frequency package of the strategy package allocated to the attached task list
- Utilise frequency codes from Appendix 2
- Use the relevant Discipline code from Appendix 3

- Use the relevant Activity code from Appendix 4
- Describe the task in detail with a maximum of 40 characters

Sequence in which information shall appear:

- Frequency Code: e.g. 3M (3 monthly)
- Discipline Code e.g. MECH
- Activity Code e.g. INSP (Inspect)
- Object Identifier e.g. CRUSHER

Acceptable:

- 3M MECH INSP CRUSHER
- 2Y INST CALB CONVEYOR WEIGHTOMETER
- 500H LUBE SVCE MINING TRUCK

Unacceptable:

- 02M MECH INSP, CV001
- 10,000H Lube Haul Truck

4.6 Maintenance Plan Short Text Description

The maintenance plan short text is used to identify plans from a list display and the maintenance plan short text is not transferred to any object. Within the short text:

- Shall describe the maintenance plan in detail (up to 40 characters)
- Use the relevant activity code from Appendix 4
- The relevant maintenance plan short text shall not contain any frequency or discipline codes
- Can contain the Tech ID to provide more information where the plan is for a single piece of equipment

Sequence in which information shall appear:

- Activity Code e.g. INSP (Inspect)
- Object Identifier e.g. CRUSHER
- Tech ID e.g. ICS1422008 (Optional)

Acceptable:

- INSP CRUSHER ICS1422008
- CALB CONVEYOR WEIGHTOMETER
- SVCE MINING TRUCK IME7628

Unacceptable:

- 03M MECH INSP, CR002
- 8 Week, Insp Conveyor

4.7 Measuring Point Short Text Description

For the measuring point the short text will clearly identify the correct measuring point to create the relevant measurement documents. The measuring point position shall contain the relevant tag number for the object. The measuring point short text description shall:

- Shall describe the measuring point in detail (up to 40 characters)
- Utilise long text for any additional information

Sequence in which information shall appear:

- Characteristic e.g. HOURS
- Object Identifier e.g. DRILL
- Object Position e.g. LH (If there are not multiple objects then this can be removed)
- Tech ID e.g. EDL254

Acceptable:

- HOURS DRILL LH EDL254
- TONNES CONVEYOR E771CV005
- KILOMETERS LIGHT VEHICLE ELV219

Unacceptable:

- Conveyor belt tonnes CV1077
- LANDCRUISER KILOMETERS

4.8 Construction Type Short Text Description

The Construction Type short text is derived from Equipment Type, manufacturer and model numbers. It is particularly important that the Construction Type description has no reference to specific physical assets as it is a company-wide definition. The Construction Type short text description shall:

- Describe the Construction Type in detail (up to 40 characters)

Sequence in which information shall appear:

- Equipment type e.g. PU (refer to Global configuration tables)
- Manufacturer e.g. WARMAN
- Model e.g. 10/8 XU
- Extended Qualifier e.g. UNLINED PUMP

Acceptable:

- PU WARMAN 10/8 XU UNLINED
- CV BELT SANDVIK
- GR JOHN DEERE 670G

Unacceptable:

- Warman Pump
- CV1077 Conveyor belt

4.9 PM Assembly Short Text Description

The PM Assembly short text is similar to the Construction type with the difference being that the extended qualifier can contain more detail for clarity. The PM Assembly short text description shall:

- Describe the PM Assembly in detail (up to 40 characters)

Sequence in which information shall appear:

- Equipment type e.g. PU (refer to Global configuration tables)
- Manufacturer e.g. WARMAN
- Model e.g. 10/8 XU
- Extended Qualifier e.g. WET END OVERHAUL

Acceptable:

- PU WARMAN 10/8 XU WET END OVERHAUL

- CV BELT SANDVIK ROLLERS
- GR JOHN DEERE 670G 500H SVC KIT

Unacceptable:

- Warman Pump
- CV1077 Conveyor belt

4.10 Work Centre Short Text Description

4.10.1 Work Centre (002 – Person)

The work centre short text is used to identify the skill type of the work crew and the primary location of this work crew. Within the short text:

- The description of the work crew's primary location will be provided i.e. Location, Department
- The skill type from the Work Crew Trade Description from the configuration tables shall be used, i.e. Discipline, Trade (Resource Work Centre only)
- The relevant work centre descriptions shall not contain individuals' names
- Refer to Region specific configuration tables for naming standrads

The work centre short text description shall be described in detail using a maximum of 40 characters.

Sequence in which information shall appear:

- Primary location e.g. MICO
- Department e.g. UG
- Discipline e.g. MTCE
- Work Crew Trade Description e.g. FITTER (Resource Work Centre only)

Acceptable (Main Work Centre):

- MICO UG MTCE

Acceptable (Resource Work Centre):

- MICO UG MTCE - FITTER

Unacceptable:

- Main Workshop Mechanical
- HV Workshop Electrical

4.10.2 Work Centre (001 – Machine)

The work centre short text is used to identify the unique machine relevant to the Equipment work centre. Within the short text;

- The Tech Id of the Equipment work centre resource will be provided, i.e. IME5752
- The Equipment type Description from Segment 3 in the Region specific configuration tables shall be used, i.e. Forklift
- The Size of the Equipment will be defined, i.e. 2 TONNE

The work centre short text description shall be described in detail using a maximum of 40 characters.

Sequence in which information shall appear:

- Tech Id e.g. IME5725
- Equipment Type e.g. FORKLIFT
- Size/Characteristic e.g. 40 FT, WATERJET

Acceptable (Equipment Work Centre):

- IME5725 FORKLIFT 40FT
- IEWSSAWJ01 PROFILE CUTTER WATERJET

Unacceptable:

- ZINC CONCENTRATOR FORKLIFT
- Profile Cutter 1

4.11 BOM Item Text 1 and 2

The BOM item texts 1 and 2 are used to provide further definition or description of BOM items in terms that are familiar to and recognised by maintenance practitioners. As these fields are primarily used to enhance existing short text or introduce colloquial terms there will be no restrictions other than the common requirements across all master data objects.

Appendices

Appendix 1 - Restricted Characters

The following characters shall not be used in any short text description.

Character	Description
_	Underscore
*	Star
?	Question mark
^	Upper char
|	Border char
~	Tilde
=	Equals
<	Less than
>	Greater than
©	Copyright symbols or similar
÷	Division
!	Exclamation mark
\	Back slash
@	Alpha

Appendix 2 - Frequency codes

Code	Description
H	Hours
D	Days
W	Weeks
M	Months
Y	Years
T	Tonnes (Metric)
TON	Ton (US Short Ton)
KM	Kilometres
MI	Miles
C	Cycles
KWH	Kilowatt Hours
L	Litres

Appendix 3 – Discipline Codes

Discipline Code	Discipline Description
3RD PRTY	3rd Party
ADMIN	Administration
CON MON	Condition Monitoring
ENG	Engineer
ELECT	Electrical
GEO	Geotechnical
INST	Instrumentation
LUBE	Lubrication
MECH	Mechanical
OPS	Operations
REFRIG	Refrigeration
SAFETY	Safety
SERV	Services
SURV	Survey
STRUCT	Structural

Appendix 4 - Activity Code Abbreviations

Activity Code	Activity Description
ADMIN	Administrative
ALIGN	Alignment / Adjustment
ANLYS	Analyse
ASSDIS	Assemble / Dismantle
ASSOP	Assistance to Operations
AUDIT	Audit
BAL	Balancing
BYPASS	Bypass
CAST	Cast
CAL	Calibration
CBM	Condition Based Monitoring
CHNG	Component Change Out
CLEAN	Cleaning / Housekeeping
CONDIS	Connect / Disconnect
CRTCTL	Critical Control
DISCARD	Discard
DRILL	Drill
DRNREF	Drain and Refill
ENV	Environmental
FAB	Fabrication / Construction
FILT	Filter
GRIND	Grind
INDCLN	Industrial Clean
INFR	Infra-red Inspection
INSP	Inspection
INST	Installation

INSCE	Insurance
LUBE	Lubrication
MATHDL	Material Handling
MEAS	Measure / Measurement
MNRMT	Minor Maintenance
MOD	Modification / Improvement
MONTR	Monitor
MTGS	Meetings
NDT	Non Destructive Testing
OVRHL	Overhaul
PST CNTRL	Pest Control
PREDMT	Predictive Maintenance
PREVMT	Preventative Maintenance
REFURB	Refurbishment
REG	Statutory / Legislation (Regulatory)
REL	Relocate
REM	Remove
REP	Repairs
ROT	Rotate
RPL	Replace
SAFETY	Safety
SAMPLE	Sample
SERV	Service
STINSP	Structural Inspecton
TEST	Testing
THERMO	Thermography
TRAIN	Training
TRBLSH	Troubleshooting
VIBE	Vibration Analysis
WARR	Warranty

5. STAKEHOLDER SIGNOFF

Stakeholder Name	Role	Contact	Stakeholder Sign-off
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